

Constant-mode transfer and compactness criteria for finite Weil characteristics

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Abstract. Restricting the Weil quadratic form to a finite Fourier space on an interval of length L gives a Hermitian matrix. When its least eigenvalue E is simple with an even eigenvector, the construction of Connes, Consani and Moscovici attaches to that eigenvector an entire characteristic with only real zeros. We show that a uniform bound on one quantity supplies both compactness of these characteristics as $L \rightarrow \infty$ and retention of the zeros of $\Xi(z) = \xi(\frac{1}{2} + iz)$. That quantity is the complete characteristic mean, a smoothed count of all zero pairs of the characteristic, including those of the untouched Fourier lattice. For finite Hermitian Fourier matrices with a rank-two displacement relation, a mean bound makes the energy above E of the ground state, smoothed by a sufficiently wide heat-difference filter, at most $CL(H_{00} - E)e^{-cL^2}$, where H_{00} is the constant-mode diagonal entry. The estimate is uniform in the Fourier cutoff and needs no spectral-gap bound; its proof transfers every entire filter of the ground state to the constant mode. For the arithmetic matrices, $H_{00} - E$ grows at most like $Le^{\frac{L}{2}}$, and the mean bound forces the cutoff to grow at least like L^2 . Consequently, if the spectral conditions and a uniform mean bound hold along $L = \log x$ for a sequence of integers $x \rightarrow \infty$, the characteristics normalized at $\frac{i}{4}$ have nonzero entire subsequential limits with only real zeros, and every zero of Ξ is a zero of each limit. Such a sequence would imply the Riemann hypothesis. Its existence and the mean bound remain open, and nonarithmetic examples show that control of the constant mode alone cannot bound the mean.

1. Introduction

The Weil explicit formula relates an arithmetic quadratic form, built from prime powers and the archimedean completion, to a sum of evaluations at the zeros of the completed ζ function. We use the conventions

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right). \quad (1)$$

Restricting the form to functions on a finite interval and then to a finite Fourier space gives a Hermitian matrix. The arithmetic and zero-side expressions of the form are recalled in [1, §3, (3.1)–(3.6)]. These finite restrictions provide a spectral construction in which the arithmetic coefficients remain explicit.

Connes, Consani and Moscovici associate an entire Fourier characteristic to the least eigenvector of such a matrix. When the full least eigenvalue is simple and its eigenvector is even, their construction gives a characteristic with only real zeros [1, Definition 5.3, Lemma 5.4 and Theorem 5.10]. After normalization at the fixed imaginary anchor $\frac{i}{4}$, write this characteristic as C . Its product contains both the zeros of a finite corrected derivative and the untouched zeros of the Fourier lattice.

The moving-limit question requires more than this finite real-zero theorem. Normalization at one point need not make a family of entire functions locally bounded. Even when a nonzero entire cluster exists, one must still establish its relation to the zeros of Ξ . Thus compactness

of the normalized characteristics and retention of the ζ zero values are distinct parts of the limiting argument.

We prove that a uniform bound for one quantity, the complete characteristic mean μ , supplies both parts on the selected families considered here. This mean is a soft count of zero pairs at the anchor scale, including the untouched lattice: nearby zeros contribute substantially, distant zeros contribute little. Its bound gives local compactness, while the anchor normalization makes each resulting cluster nonzero. The same bound makes a smoothed ground-state residual Gaussian-small in the interval length. For the arithmetic matrices, that residual estimate implies that every zero of Ξ is a zero of every characteristic cluster. This conclusion is conditional on one unbounded family satisfying the full simple-even selection hypothesis and the uniform mean bound.

Here is the finite estimate. Let H be the complete Fourier matrix on an interval of length L , let E be its full least eigenvalue, and put $K = H - EI \geq 0$. The anchor-normalized ground vector h satisfies $Kh = 0$. Smoothing by the fixed heat difference $B_s = e^{-(s+1)D^2} - e^{-(s+2)D^2}$ need not preserve that equation, so we measure its residual by the shifted energy $\varepsilon = \langle B_s h, KB_s h \rangle$. We prove

$$\varepsilon \leq C_{M,s} L(H_{00} - E)e^{-d_{M,s}L^2}, \quad \mu \leq M, \quad s > 16M, \quad (2)$$

with positive constants independent of the Fourier cutoff. The explicit exponent is given in (39). Positive-form Cauchy–Schwarz gives

$$|\langle p, KB_s h \rangle|^2 \leq \langle p, Kp \rangle \varepsilon. \quad (3)$$

Thus the estimate controls the smoothed equation against test vectors whose shifted energies can be bounded; Appendix A supplies those test estimates. No lower bound on the nonzero spectrum of K enters.

The factor $H_{00} - E$ is the shifted energy of the unit constant Fourier mode. Corrected differentiation transfers every entire filter of the ground vector to that mode. On the quotient by the ground line, the filtered energy becomes a spectral integral with total measure $H_{00} - E$. Its divided difference is a sine transform of the exterior tail of a whole-line smoothing. Pairing all periodization images removes the apparent Fourier-lattice denominator; bounded mean then gives the Gaussian tail estimate. The proof estimates the complete filtered energy and preserves cancellation between its arithmetic components.

For the arithmetic form we also prove $H_{00} - E \leq C(1 + L)e^{\frac{L}{2}}$, so the filtered energy tends to zero on a mean-bounded selected family. The Fourier resolution needed for the moving current argument is itself a consequence of the complete mean bound: the untouched lattice forces N to grow at least quadratically in L . Translated Gaussian tests then have Fourier projection errors small enough to dominate the arithmetic growth. Their pairings determine the whole-line current in its actual exponential-growth class. Section 7 and Appendix A prove zero retention for arbitrary positive integer cutoffs on the same mean-bounded selected family, with $L = \log x$ and $x \rightarrow \infty$. Existence of that family and its mean bound remain the two arithmetic premises.

Bombieri’s fixed-interval variational study of the Weil functional includes finite Fourier truncations [2, abstract, p. 183]. The finite characteristic and corrected-derivative geometry used here come from [1]; see also [3, §5]. The results here add the all-filter constant-mode identity, its boundary-tail estimate and the conditional current transfer. Suzuki’s finite-interval construction uses a Friedrichs realization and a metric shifted strictly below the spectral bottom [4, Theorems 1.1 and 1.5]. Our shifted form instead vanishes on the selected ground line, which is removed in forming the quotient.

Sections 2–5 establish the finite transfer and Gaussian estimate. Section 6 proves the arithmetic constant-mode bound. Section 7 and Appendix A give the moving-family argument. Section 8 explains, by a nonarithmetic example, why the weighted constant-mode observation does not itself bound the unweighted root count.

2. Fourier matrices and selected characteristics

All inner products are conjugate-linear in their first argument. Let $L > 0$ and let $N \geq 1$ be an integer. Put $I = [-b, b]$, $b = \frac{L}{2}$, and let V_N be the span in $L^2(I)$ of

$$e_j(t) = L^{-\frac{1}{2}} e^{i\xi_j t}, \quad \xi_j = 2\pi \frac{j}{L}, \quad -N \leq j \leq N. \quad (4)$$

We identify vectors and their Fourier polynomials. The Fourier generator and endpoint vector are

$$D = \text{diag}(\xi_j), \quad \delta_j = \frac{(-1)^j}{\sqrt{L}}, \quad \delta^* f = f(b). \quad (5)$$

The unit constant mode is e_0 ; in particular $\delta^* e_0 = L^{-\frac{1}{2}}$. Reflection sends the coefficient at j to that at $-j$.

Assume that H is real symmetric, commutes with reflection, and satisfies

$$[D, H] = b\delta^* - \delta b^*, \quad b \text{ is odd}. \quad (6)$$

Assume its full least eigenvalue E is simple and has an even eigenvector. These are the finite selection hypotheses throughout. Set $K = H - EI \geq 0$. Its kernel is the selected line. If an even ground vector v had $\delta^* v = 0$, (6) would give $KDv = 0$. The odd vector Dv must then vanish. A constant vector with zero endpoint is zero, a contradiction. We can therefore choose the real endpoint normalization $\delta^* v = 1$.

For any vector f , write

$$\ell_z(f) = \int_I f(t) e^{-izt} dt. \quad (7)$$

Direct integration of each Fourier mode gives the entire interpolation identity

$$\ell_z(f) = 2 \sin(bz) \delta^* (z - D)^{-1} f, \quad (8)$$

with removable values at retained frequencies. Let $\alpha = \frac{1}{4}$. The imaginary anchor $a = \ell_{i\alpha}(v)$ is positive, as the product below shows. Define

$$h = \frac{v}{a}, \quad C(z) = \ell_z(h), \quad C(i\alpha) = 1. \quad (9)$$

Neither normalization entails a uniform lower bound on a physical trace.

We next record the characteristic facts needed for the mean estimate. Their Hilbert-space input is the self-adjointness of the corrected derivative $D_c = D - Dv\delta^*$ on the quotient by the ground line. This finite construction and the resulting real-zero characteristic are due to Connes, Consani and Moscovici; we give the needed algebra to fix the normalization. Section 3 constructs that realization directly from (6). Using its self-adjointness, the following calculation verifies the nonzero simple spectrum and the complete product. It has $2N$ nonzero eigenvalues $\pm\eta_k$, with $\eta_k > 0$.

To check that zero is absent, first observe that $\ker D_c = \mathbb{C}v$: from $D_c u = 0$ one gets $u = v\delta^* u + ce_0$, and application of δ^* gives $c = 0$. A zero eigenvector in the quotient is mapped

by D_c injectively into $\mathbb{C}v$, so that quotient kernel has dimension at most one. Reflection interchanges the signs of D_c ; the even and odd quotient spaces both have dimension N . The self-adjoint quotient has off-diagonal square blocks, hence an even-dimensional kernel. Its kernel is therefore zero. Every nonzero eigenvalue is simple: on its eigenspace the functional δ^* is injective, since $\delta^*u = 0$ would imply $(D - \lambda)u = 0$, and every Fourier basis vector has nonzero endpoint. Quotient self-adjointness supplies semisimplicity.

The determinant lemma now gives

$$Q(z) := \frac{\det(z - D_c)}{\det(z - D)} = z\delta^*(z - D)^{-1}v = \prod_{k=1}^N \frac{z^2 - \eta_k^2}{z^2 - \xi_k^2}. \quad (10)$$

In particular, $a = 2 \sinh(\alpha b) \frac{Q(i\alpha)}{\alpha} > 0$. Together with the sine product and (8), this yields

$$C(z) = \prod_{k=1}^N \frac{\eta_k^2 - z^2}{\eta_k^2 + \alpha^2} \prod_{n>N} \frac{\xi_n^2 - z^2}{\xi_n^2 + \alpha^2}. \quad (11)$$

Coincident factors are interpreted by cancellation in the polynomial determinant identity. A corrected root coinciding with an untouched lattice root is counted twice. The infinite product converges locally uniformly. Although h is a finite Fourier polynomial, its transform C is an entire function with infinitely many zeros: for every $|n| > N$, Fourier orthogonality gives $C(\xi_n) = 0$. These unretained lattice zeros are exactly the factors in the second product.

A zero pair $\pm\lambda$ contributes the weight $\frac{\alpha^2}{\lambda^2 + \alpha^2}$, which decreases from one to zero as its distance from the origin increases. The sum is therefore a smoothed count of zero pairs at scale α . Writing each product factor as $1 - r_l - r_l \frac{z^2}{\alpha^2}$, with $0 < r_l < 1$, we define the characteristic mean

$$\mu = \sum_l r_l = \sum_{k=1}^N \frac{\alpha^2}{\eta_k^2 + \alpha^2} + \sum_{n>N} \frac{\alpha^2}{\xi_n^2 + \alpha^2}. \quad (12)$$

For $N \geq 1$ the last sum is at most $\alpha^2 \frac{L^2}{4\pi^2 N}$. It is finite for each parameter pair (L, N) ; a bound uniform in a moving family is a separate condition. Because the untouched terms are positive, such a bound also restricts the cutoff: Appendix A.2 proves $N \geq \frac{L^2}{128\pi^2 M}$ for large L whenever $\mu \leq M$.

3. Corrected differentiation and the constant mode

The shifted form has exactly one null direction, the selected ground line. Removing it makes the form a positive inner product to which the spectral theorem applies. The correction in D_c makes differentiation compatible with this quotient: it kills v and restores symmetry for the shifted form. Indeed, applying (6) to v , using parity, gives $\mathbf{b} = -KDv$. Consequently

$$D_c v = 0, \quad KD_c = D_c^* K. \quad (13)$$

We realize this quotient inside the original coefficient space by its square-root map. Let $S = K^{\frac{1}{2}}$, $\mathcal{E} = \text{Ran } S = (\mathbb{C}v)^\perp$, and let $S^\dagger$ denote the finite Moore–Penrose inverse. Define

$$A = SD_c S^\dagger|_{\mathcal{E}}. \quad (14)$$

If $P = S^\dagger S$, then $D_c(I - P) = 0$, so

$$AS = SD_c. \quad (15)$$

For $Su, Sw \in \mathcal{E}$, (13) gives

$$\langle Su, ASw \rangle = \langle u, KD_c w \rangle = \langle D_c u, Kw \rangle = \langle ASu, Sw \rangle. \quad (16)$$

Thus A is self-adjoint in the inherited physical norm on $\mathcal{E}$. The map $[u] \mapsto Su$ is an isometry from the positive K -quotient to this space. No estimate of $S^\dagger$ has entered. The constant mode enters because $De_0 = 0$: only the rank-one correction acts on it. In particular,

$$D_c e_0 = -D \frac{v}{\sqrt{L}}, \quad A S e_0 = -S D \frac{v}{\sqrt{L}}. \quad (17)$$

Thus the single vector $S e_0$ encodes the corrected derivative of the selected ground. The next identity iterates this relation to express every entire filter of h through the spectral measure of that vector.

Theorem 1 (Constant-mode transfer).

For every entire function φ , define the entire divided difference

$$r_{\varphi,h}(z) = \sum_{j=-N}^N \bar{\delta}_j h_j \frac{\varphi(\xi_j) - \varphi(z)}{z - \xi_j}. \quad (18)$$

Then

$$\varphi(D)h = \varphi(0)h + \sqrt{L} D_c r_{\varphi,h}(D_c) e_0. \quad (19)$$

Let ν_0 be the spectral measure of A for $S e_0$. Its mass and the filtered energy are

$$\nu_0(\mathbb{R}) = H_{00} - E, \quad (20)$$

$$\langle \varphi(D)h, K \varphi(D)h \rangle = L \int_{\mathbb{R}} |\lambda r_{\varphi,h}(\lambda)|^2 d\nu_0(\lambda). \quad (21)$$

Proof. At $z = \xi_j$ the corresponding summand in (18) has value $-\bar{\delta}_j h_j \varphi'(\xi_j)$. Thus there is no pole at a Fourier or corrected frequency. For $\varphi(z) = z^n$, $n \geq 1$,

$$r_{n,h}(z) = - \sum_{k=0}^{n-1} z^k \delta^* D^{n-1-k} h. \quad (22)$$

The ordered telescoping identity

$$D^n - D_c^n = \sum_{k=0}^{n-1} D_c^k (D - D_c) D^{n-1-k} \quad (23)$$

and $D_c h = 0$ express $D^n h$ as $\sum_k D_c^k D v \delta^* D^{n-1-k} h$. Equation (17) gives (19). The constant monomial has zero divided difference and contributes $\varphi(0)h$.

For an entire power series, choose for this fixed finite section a radius larger than both matrix norms. The divided-difference series is dominated by a constant times $\sum_n |a_n| n R^{n-1}$ and converges in matrix norm. This proves the identity for entire functions without assumptions about simple corrected roots. By (15),

$$S \varphi(D)h = \sqrt{L} A r_{\varphi,h}(A) S e_0. \quad (24)$$

The spectral theorem proves (21), and $\|S e_0\|^2 = \langle e_0, K e_0 \rangle = H_{00} - E$ gives the mass. $\square$

In particular, whenever the real supremum is finite,

$$\langle \varphi(D)h, K \varphi(D)h \rangle \leq L(H_{00} - E) \sup_{\lambda \in \mathbb{R}} |\lambda r_{\varphi,h}(\lambda)|^2. \quad (25)$$

This estimate is in the original shifted form. It contains neither a small-gap denominator nor a comparison between physical and shifted norms. If $Se_0 = 0$, the selected ground is constant and every filtered shifted energy vanishes. If the chosen eigenvalue were not the full minimum, K could be indefinite and the positive spectral-measure argument would no longer apply.

4. Divided differences as boundary tails

Fix the heat-difference filter

$$\varphi_s(z) = e^{-(s+1)z^2} - e^{-(s+2)z^2}, \quad B_s = \varphi_s(D). \quad (26)$$

Our Fourier convention is $\hat{f}(z) = \int_{\mathbb{R}} f(t) e^{-izt} dt$, with inverse factor $\frac{1}{2\pi}$. Let

$$U = \mathcal{F}^{-1}(\varphi_s C), \quad \tau(t) = \sum_{k \geq 0} U(b + t + kL), \quad t \geq 0. \quad (27)$$

For each fixed finite Fourier space, U is the difference of two Gaussian convolutions of the zero extension of h . It is even and has Gaussian tails, as do its derivatives. Uniformity under a mean bound will be proved in Section 5.

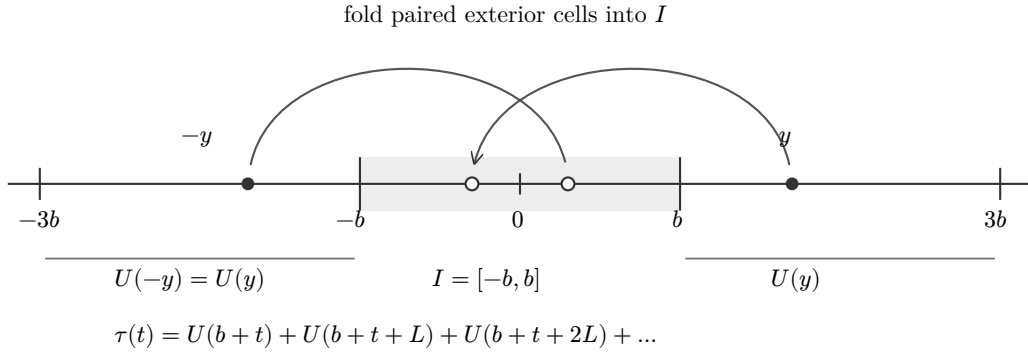


Figure 1: The first pair of exterior cells, with $L = 2b$. Periodization sends y to $y - L$ and $-y$ to $-y + L$ (open circles). Pairing their cosine contributions and summing all exterior cells produces the single tail τ in the sine representation.

The coefficient of $e^{i\xi_j t}$ in the L -periodization of U is $\varphi_s(\xi_j) \frac{C(\xi_j)}{L}$. For retained frequencies $C(\xi_j) = \sqrt{L}h_j$; for every unretained integer frequency it is zero. Therefore

$$B_s h = \text{per}_L U|_I. \quad (28)$$

No Fourier remainder occurs in this formula.

Proposition 2 (Sine-tail representation).

For every complex z , the divided difference of (18) satisfies

$$r_s(z) = 2 \int_0^\infty \tau(t) \sin(zt) dt. \quad (29)$$

For real λ , including zero,

$$\lambda r_s(\lambda) = 2\tau(0) + 2 \int_0^\infty \tau'(t) \cos(\lambda t) dt. \quad (30)$$

Consequently

$$\langle B_s h, K B_s h \rangle \leq 4L(H_{00} - E) \left(|\tau(0)| + \|\tau'\|_1 \right)^2. \quad (31)$$

Proof. By (8) and (28),

$$2 \sin(bz) r_s(z) = \int_I \text{per}_L U(t) e^{-izt} dt - \int_{\mathbb{R}} U(t) e^{-izt} dt. \quad (32)$$

Pair positive and negative exterior cells. On $y \in [(2m-1)b, (2m+1)b]$, $m \geq 1$, their contribution is

$$2U(y) \{ \cos(z(y - mL)) - \cos(zy) \} dy. \quad (33)$$

The finite telescoping identity

$$\cos(z(y - mL)) - \cos(zy) = 2 \sin(bz) \sum_{k=0}^{m-1} \sin(z(y - (2k+1)b)) \quad (34)$$

holds for all complex z . For a fixed k , the cells with $m \geq k+1$ cover $y \geq (2k+1)b$. Substitute $t = y - (2k+1)b$ and sum. Gaussian decay dominates the cell count and the exponential factor on each compact z -set; all interchanges converge absolutely and locally uniformly. Cancelling $\sin(bz)$ off its zeros gives (29). Both sides are entire, so the identity includes the entire lattice.

Since $\tau, \tau' \in L^1$ and $\tau(\infty) = 0$, integration by parts gives (30) when $\lambda \neq 0$. At zero its right side is $2\tau(0) + 2 \int \tau' = 0$, agreeing with the left side. Taking the supremum and applying (25) proves the estimate. $\square$

This calculation removes the apparent small denominator in

$$r_s(z) = \frac{\ell_z(B_s h) - \varphi_s(z) C(z)}{2 \sin(bz)}. \quad (35)$$

The exterior tail, rather than the distance to a Fourier lattice point, controls the filtered variation. All images are part of that tail.

5. Characteristic mean and Gaussian variation

The boundary-tail estimate reduces the energy problem to decay of a whole-line smoothing outside $[-\frac{L}{2}, \frac{L}{2}]$. A bound on the characteristic mean supplies that decay by controlling growth of the entire transform. From (11) and $|1 - r - r \frac{z^2}{\alpha^2}| \leq 1 + r|z \frac{1}{\alpha^2}|^2$, the complete mean gives

$$|C(z)| \leq \exp(16\mu|z|^2). \quad (36)$$

Logarithmic differentiation at the anchor gives

$$\mu = \left(i \frac{\alpha}{2}\right) C'(i\alpha). \quad (37)$$

At $\beta = \frac{3}{16}$, set $F_0 = C(i\beta)$. Each factor equals $1 - 7 \frac{r_i}{16}$. Convexity of $-\log(1 - 7 \frac{r}{16})$ on $[0, 1]$ gives

$$\left(\frac{7}{16}\right) \mu \leq -\log F_0 \leq \log\left(\frac{16}{9}\right) \mu. \quad (38)$$

Thus a uniform mean bound is equivalent to a positive lower bound at this fixed subanchor.

Theorem 3 (Gaussian filtered variation).

Assume $L \geq 1$, $\mu \leq M$, and fix $s > 16M$. For the filter (26),

$$0 \leq \varepsilon := \langle B_s h, K B_s h \rangle \leq C_{M,s} L(H_{00} - E) \exp(-d_{M,s} L^2), \quad (39)$$

where one may take $d_{M,s} = \frac{1}{16(s+2+16M)}$. The constant is independent of the finite Fourier cutoff.

Proof. We first turn the entire growth estimate into Gaussian bounds for the smoothed function and its derivative. Summing those bounds over the exterior cells then controls the sine-tail formula from Section 4.

Put $q = s + 1 - 16M > 0$ and $p = s + 2 + 16M$. For $a = s + 1$ or $s + 2$ and $z = \xi + iy$, (36) implies

$$|e^{-az^2}C(z)| \leq \exp(-(a - 16M)\xi^2 + (a + 16M)y^2). \quad (40)$$

Shift the inverse Fourier contour to $y = \frac{t}{2p}$. The vertical sides tend to zero by the positive Gaussian margin q , also after inserting iz for a derivative. Integrating the two heat terms yields

$$|U(t)| \leq (\pi q)^{-\frac{1}{2}} e^{-\frac{t^2}{4p}}, \quad (41)$$

$$|U'(t)| \leq \pi^{-1} \left(q^{-1} + |t| \frac{1}{2p} \sqrt{\frac{\pi}{q}} \right) e^{-\frac{t^2}{4p}}. \quad (42)$$

In particular, with $c = \frac{1}{8p}$, $|U(t)| + |U'(t)| \leq C_{M,s} e^{-ct^2}$. This directly estimates the derivative under the contour integral.

For $L \geq 1$,

$$(b + t + kL)^2 \geq b^2 + t^2 + k^2 L^2, \quad (43)$$

so termwise differentiation in (27) and integration give

$$|\tau(0)| + \|\tau'\|_1 \leq C_{M,s} \left(1 + \int_0^\infty e^{-ct^2} dt \right) \sum_{k \geq 0} e^{-ck^2} e^{-cb^2}. \quad (44)$$

Its square is bounded by a constant times $e^{-\frac{L^2}{16p}}$. Now apply (31). $\square$

The filter remains fixed along a mean-bounded family. The same product estimate supplies local boundedness of all its characteristics on the entire plane. Every locally uniform cluster has value one at $\frac{i}{4}$, and hence is nonzero. The trace formula

$$\mu = \frac{1}{2} \operatorname{tr}_{\mathcal{E}} \left(\alpha^2 (A^2 + \alpha^2 I)^{-1} \right) + \mu_{\text{lat}} \quad (45)$$

will distinguish this compactness condition from the weighted spectral observation in (21).

6. The finite Weil form and its constant-mode cost

We now use the arithmetic matrix of the completed-operator companion [5, §4, Theorem “The complete arithmetic Fourier matrix”], with diagonal entries defined by confluent divided differences. Write f^0 for zero extension from I . Set

$$a_0 = \psi\left(\frac{1}{4}\right) - \log \pi, \quad \lambda_k = 2k + \frac{1}{2}, \quad \rho(q) = \sum_{k \geq 0} e^{-\lambda_k q}. \quad (46)$$

Here ψ is the digamma function. For $q > 0$, let T_q be the killed shift on I ; the two orientations give adjoints. The complete quadratic form is

$$\begin{aligned}
W_L(f, f) &= a_0 \|f\|^2 + \int_0^\infty \rho(q) \|f^0(\cdot + q) - f^0\|^2 dq \\
&\quad - 2 \sum_{1 < n < e^L} \frac{\Lambda(n)}{\sqrt{n}} \Re \langle f, T_{\log n} f \rangle \\
&\quad + 2 \left| \left\langle \cosh\left(\frac{\cdot}{2}\right), f \right\rangle \right|^2 - 2 \left| \left\langle \sinh\left(\frac{\cdot}{2}\right), f \right\rangle \right|^2.
\end{aligned} \tag{47}$$

Its polarization uses conjugate-linear-first inner products, and H is its restriction to V_N . Strict prime cutoffs are retained. A shift of exactly L vanishes in L^2 . The logarithmic singularity $\rho(q) = O\left(\frac{1}{q}\right)$ is harmless on finite Fourier polynomials: the interior translation difference is $O(q^2)$ in squared norm and the endpoint strips contribute $O(q)$. Zero extension is not being treated as an $H^1(\mathbb{R})$ function.

This form gives the displacement in (6). For clarity, the matrix can also be specified directly. Let

$$a_\Gamma(\xi) = \Re \psi\left(\frac{1}{4} + i\frac{\xi}{2}\right) - \log \pi, \tag{48}$$

$$\begin{aligned}
j_L(\xi) &= - \sum_{1 < n < e^L} \frac{\Lambda(n)}{\sqrt{n}} \sin(\xi(L - \log n)) + \left(e^{\frac{L}{2}} - 1\right) \frac{\xi}{\xi^2 + \frac{1}{4}} \\
&\quad + \xi \sum_{k \geq 1} \frac{1 - e^{-\lambda_k L}}{\lambda_k^2 + \xi^2}.
\end{aligned} \tag{49}$$

Then

$$H_{jk} = a_\Gamma(\xi_j) \delta_{jk} + 2 \frac{(-1)^{j+k}}{L} j_L[\xi_j, \xi_k], \tag{50}$$

where the confluent divided difference is $j_L[\xi_j, \xi_j] = j'_L(\xi_j)$. Thus $H_{jj} = a_\Gamma(\xi_j) + \frac{2j'_L(\xi_j)}{L}$.

The coordinate relation with that companion is explicit. It uses $U_{j(t)} = L^{-\frac{1}{2}} e^{i\xi_j t}$ on $(0, L)$; centering by $\mathcal{T}f(t) = f\left(t + \frac{L}{2}\right)$ gives $\mathcal{T}U_j = (-1)^j e_j$. With H^I and $\mathbf{a}_{e^L}$ denoting its matrix and interpolation function, the formulas above give

$$j_L = \mathbf{a}_{e^L}, \quad H_{jk} = (-1)^{j+k} H_{jk}^I. \tag{51}$$

This includes the derivative values on the diagonal. Its endpoint-one ground vector ξ_N becomes $v = \mathcal{T}\xi_N$. Its characteristic $\Theta_{L,N}^I$ is $L^{-1}\ell_z(v)$, so the anchor normalization is

$$C_{L,N}(z) = \frac{\Theta_{L,N}^I(z)}{\Theta_{L,N}^I\left(\frac{i}{4}\right)}. \tag{52}$$

These are the same selected finite characteristic and physical form in centered coordinates [5, §§4–5, centered-coordinate formula and Theorem “Finite boundary characteristic”]. The equality is a unitary change of physical Fourier coordinates; the companion’s source-Gram derivative represents the form on completed outputs and is a different coordinate representation. The finite transfer in Sections 3–5 applies to the general displacement setting; the cost estimate below and the moving theorem use this arithmetic specialization.

Proposition 4 (Constant-mode cost).

For every finite section, $L \geq 1$,

$$0 \leq H_{00} - E \leq 2 \sum_{k \geq 0} \lambda_k^{-2} + \left(4L + 1 + \frac{8}{L}\right) e^{\frac{L}{2}} \leq C(1 + L) e^{\frac{L}{2}}. \quad (53)$$

No selection assumption is required for this bound on the full least eigenvalue.

Proof. Killed shifts have norm at most one. Using $\Lambda(n) \leq \log n \leq L$ and $\sum_{n \leq x} n^{-\frac{1}{2}} \leq 2\sqrt{x}$, the prime part is bounded below by $-4L e^{\frac{L}{2}} \|f\|^2$. The negative polar part has norm

$$2 \left\| \sinh\left(\frac{\cdot}{2}\right) \right\|_{L^2(I)}^2 = 2 \sinh\left(\frac{L}{2}\right) - L \leq e^{\frac{L}{2}}. \quad (54)$$

The Γ form is bounded below by $a_0 \|f\|^2$. Hence $E \geq a_0 - (4L + 1) e^{\frac{L}{2}}$.

For the unit constant e_0 , the squared translation difference is $2 \min(\frac{q}{L}, 1)$. Positive summation and integration therefore give

$$\Gamma_{00} = a_0 + \frac{2}{L} \sum_{k \geq 0} \frac{1 - e^{-\lambda_k L}}{\lambda_k^2}. \quad (55)$$

The constant prime diagonal is $-2 \sum_{1 < n < e^L} \Lambda(n) n^{-\frac{1}{2}} \left(1 - \frac{\log(n)}{L}\right) \leq 0$. The polar diagonal is $32 \sinh^2 \frac{L}{4} \leq 8 e^{\frac{L}{2}}$. Combining these estimates and $E \leq H_{00}$ proves the assertion. $\square$

Combining (53) with (39) gives, for every selected finite matrix with $\mu \leq M$,

$$\varepsilon \leq C_{M,s} L(1 + L) \exp\left(\frac{L}{2} - \frac{L^2}{16(s + 2 + 16M)}\right). \quad (56)$$

This filtered-energy bound holds for arbitrary finite N . Appendix A shows that the complete mean bound also supplies the Fourier resolution needed to pass the finite equation to the global current.

7. Compactness and zeros of the limit

Write $\xi(s) = \frac{1}{2} s(s-1) \pi^{-\frac{s}{2}} \Gamma(\frac{s}{2}) \zeta(s)$ and $\Xi(z) = \xi(\frac{1}{2} + iz)$. The global Weil form W is obtained from (47) on whole-line tests with sufficient decay, with all integer prime shifts. Its polarized explicit formula, in this normalization, is [1, §3, (3.1)–(3.6)]

$$W(f, g) = \sum_{\Xi(\gamma)=0} \overline{\hat{f}(\bar{\gamma})} \hat{g}(\gamma), \quad (57)$$

where zeros are counted with multiplicity. We use this formula only for the smooth rapidly decreasing test classes justified in the appendix.

The finite variation estimate gives a current equation for a smoothed entire limit. A nonzero anchor moment first prevents collapse of the smoothed states and bounds their ground energies. The complete mean forces enough Fourier resolution to pass the small residual to translated Gaussian tests on the line. Those tests determine the physical current in its exponential-growth class. θ derivatives whose transforms vanish at every zero of Ξ then eliminate the limiting energy by density. Finally, dividing the θ transform by one conjugated zero-pair factor isolates that pair in the second slot of the explicit formula. Appendix A proves these steps on their required domains.

Theorem 5 (A conditional moving-family criterion).

Let x_j be a sequence of integers $x_j \geq 2$ tending to infinity, $L_j = \log x_j$, and let $N_j \geq 1$ be arbitrary integers. Suppose each complete matrix (50) has a simple full least eigenvalue with an even eigenvector. Normalize its characteristic at $\frac{i}{4}$, and assume $\mu_j \leq M < \infty$ on this same family. The complete mean then forces $N_j \geq \frac{L_j^2}{128\pi^2 M}$ for all sufficiently large j .

The characteristics form a locally bounded entire family. Every locally uniform subsequential limit F is nonzero, has only real zeros, and satisfies

$$F(\gamma) = 0 \quad \text{whenever} \quad \Xi(\gamma) = 0. \quad (58)$$

Moreover $E_j \rightarrow 0$, and the fixed filtered variation tends to zero for every $s > 16 \sup_j \mu_j$.

Proof. Local boundedness and $F(\frac{i}{4}) = 1$ follow from (36). Variation decay is (56). The resolution lemma in Appendix A gives the stated cutoff consequence from the untouched lattice alone. The norm and current lemmas then show that, after passage to an entire cluster and an energy cluster, the whole-line filtered state

$$U_F = \mathcal{F}^{-1}(\varphi_s F) \quad (59)$$

satisfies $\mathcal{H}U_F = EU_F$ pointwise. The proof first passes the finite residual to all translates of one fixed Gaussian and then identifies the smooth current by a Gaussian-weighted bilateral Laplace transform. The norm lemma bounds both signs of the finite energies before extracting that cluster.

The dense-radical lemma in Appendix A supplies even θ -derived tests that can be paired with this equation and whose span is dense in ordinary even L^2 . They give $E = 0$, because $U_F \neq 0$. The zero-isolation lemma then gives $\varphi_s(\gamma)F(\gamma) = 0$ at every zero of Ξ . The nonzero zeros of φ_s satisfy $z^2 = 2\pi ik$, $k \in \mathbb{Z} \setminus \{0\}$, and hence $|\Im z| \geq \sqrt{\pi}$. Zeros of Ξ lie in $|\Im z| \leq \frac{1}{2}$; also $\Xi(0) > 0$. Thus the filter is nonzero at every such zero, proving that every zero of Ξ is a zero of F . Since every energy cluster is zero, $E_j \rightarrow 0$. Finally, the original characteristics (11) have real zeros only; Hurwitz applied on each nonreal disk shows the same for their nonzero limit. $\square$

The theorem gives an RH-sufficient criterion: a nonzero cluster has only real zeros and contains every zero of Ξ , so all zeros of Ξ must be real. Its two moving-family premises remain open: an unbounded sequence of the stated full simple-even minima, and a uniform bound on their characteristic means. They must hold on the same sequence. We call the conclusion $\Xi(\gamma) = 0 \Rightarrow F(\gamma) = 0$ zero retention; it specifies zero values, without fixing their multiplicities or determining the entire function F uniquely.

8. Counting roots and observing the constant mode

The finite energy theorem controls an integral against ν_0 . The mean in (45) counts corrected eigenvalues without their observation weights. The distinction persists even when the constant vector sees every corrected mode.

Proposition 6 (A weighted observation with unbounded root count).

There are nonarithmetic matrices satisfying the Fourier, displacement and full simple-even selection hypotheses, with every corrected mode visible to Se_0 and $H_{00} - E = 1$, for which $\mu > 4\frac{N}{5}$. Thus the mass bound and all-filter transfer do not imply a uniform characteristic-mean bound.

Proof. Choose distinct $0 < \eta_k < \frac{\alpha}{2}$ avoiding the retained Fourier frequencies. Form

$$Q(z) = \prod_{k=1}^N \frac{z^2 - \eta_k^2}{z^2 - \xi_k^2}, \quad \delta^*(z - D)^{-1}v = \frac{Q(z)}{z}. \quad (60)$$

Residues determine a real even v . The coefficient at infinity gives $\delta^*v = 1$. The determinant lemma then makes $D_c = D - Dv\delta^*$ have simple spectrum $0, \pm\eta_1, \dots, \pm\eta_N$. Write P_λ for its algebraic spectral projections. For arbitrary positive symmetric numbers $t_\lambda = t_{-\lambda}$, define

$$K_t = \sum_{\lambda \neq 0} t_\lambda P_\lambda^* P_\lambda. \quad (61)$$

This matrix is positive with kernel $\mathbb{C}v$, commutes with reflection, and satisfies $K_t D_c = D_c^* K_t$. Expanding this last identity recovers (6) with $\mathbf{b} = -K_t Dv$. Hence $H = K_t$ has full simple-even minimum $E = 0$ and the same characteristic.

The spectral mass of its quotient at λ is

$$\nu_t(\{\lambda\}) = t_\lambda \|P_\lambda e_0\|^2. \quad (62)$$

Every norm here is nonzero. Indeed, the rank-one resolvent identity gives

$$(z - D_c)^{-1} e_0 = \frac{e_0}{z} - \delta_0 (z - D)^{-1} D \frac{v}{zQ(z)}; \quad (63)$$

at a corrected root outside the retained free spectrum its residue is nonzero. Otherwise $Dv = 0$, which would make all corrected roots free. Choose t_λ so that these masses equal $\frac{1}{2N}$. Their total is one, whereas each positive corrected root contributes more than $\frac{4}{5}$ to (12). The free lattice adds a nonnegative term. $\square$

One can additionally rescale K_t to have operator norm at most one; this preserves all roots and their visibility and makes the total observation mass at most one. These matrices do not have the literal von Mangoldt, Γ and polar entries. They show why a mean bound must use a quantitative arithmetic property beyond displacement and shifted positivity.

For the finite Weil family, the remaining compactness condition can be expressed either as bounded μ or, by (38), as a fixed positive floor for $C(3\frac{4}{16})$. The energy estimate proved here supplies the filtered part of the moving-family argument once that condition and an unbounded selected family are available.

Appendix B expresses this same complete-mean condition through the ratio of two inverse quadratic forms for explicit odd hyperbolic sources. It retains the full least energy and the untouched lattice; estimating that ratio is another form of the remaining arithmetic question.

A Current domains and the moving limit

This appendix proves the moving argument in Section 7. The complete mean first forces the Fourier resolution; Gaussian tests then pass the finite equation to the whole line. The final θ tests identify its energy and zero values. Let W be the whole-line polarization of (47). Compact smooth functions, Gaussian functions with Gaussian bounds for every derivative, and the θ -derived functions constructed below are the only global test classes needed. In particular, we do not extend the prime current as a bounded operator on ordinary L^2 .

A.1 A smoothed norm floor and bounded energies

Lemma (Smoothed norm and energy bounds). The filtered states have uniformly bounded Weil form values and L^2 norms, their norms are bounded away from zero, and the ground energies remain in a bounded real interval.

Proof. Throughout this appendix $\mu_j \leq M$ and $s > 16M$ are fixed, $w_j = B_s h_j$, and $U_j = \mathcal{F}^{-1}(\varphi_s C_j)$. Set $p_s = s + 2 + 16M$. Contour shifting as in (42), with any fixed polynomial inserted, gives $|U_j^{(k)}(t)| \leq C_k e^{-\frac{t^2}{8p_s}}$, uniformly in j . Exact periodization gives, on I_j ,

$$|w_j^{(k)}(t)| \leq C_k e^{-c_k t^2}, \quad \|w_j - U_j\|_{H^1(I_j)} + \|U_j\|_{H^1(\mathbb{R} \setminus I_j)} \leq C e^{-c L_j^2}. \quad (64)$$

The same statement holds for each fixed exponentially weighted norm after reducing c .

These bounds imply

$$|W_{L_j}(w_j, w_j)| + \|w_j\|_{H^1(I_j)}^2 \leq C. \quad (65)$$

For Γ , split the translation integral at $q = 1$. For $q \leq 1$ the interior squared difference is at most $q^2 \|w'_j\|^2$, and the two endpoint strips are at most $2q \|w_j\|_\infty^2$. Multiplication by $\rho(q) = O(\frac{1}{q})$ is integrable. For $q \geq 1$, use the uniform L^2 norm and the exponential tail of ρ . For the killed prime correlation, completing the square gives

$$|\langle w_j, T_q w_j \rangle| \leq C \int_{\mathbb{R}} e^{-ct^2 - c(t+q)^2} dt \leq C e^{-c' q^2}. \quad (66)$$

Thus its absolute sum is bounded by $C \sum_{n \geq 2} \Lambda(n) n^{-\frac{1}{2}} e^{-c'(\log n)^2} < \infty$. Both polar moments are bounded by the Gaussian exponential moments. This proves (65) for the complete form with its killed overlaps.

There is also a lower norm bound. The anchor gives the ordinary convergent moment

$$\int_{\mathbb{R}} U_j(t) e^{\alpha t} dt = \varphi_s(i\alpha) = e^{(s+1)\alpha^2} - e^{(s+2)\alpha^2} \neq 0. \quad (67)$$

The uniform weighted Gaussian tails allow one fixed compact interval to contain at least half the absolute value of this moment. Cauchy–Schwarz there gives $\|U_j\|_2 \geq c > 0$, hence by (64) also $\|w_j\| \geq \frac{c}{2}$ for large j . Therefore, once $\varepsilon_j \leq 1$,

$$E_j = \frac{W_{L_j}(w_j, w_j) - \varepsilon_j}{\|w_j\|^2} \quad (68)$$

bounds both signs of E_j uniformly. The fixed nonzero anchor moment supplies the norm floor after smoothing; no uniform lower bound for the unsmoothed trace is required.

A.2 The complete mean forces Fourier resolution

Lemma (Resolution from the lattice tail). Suppose $\mu \leq M$, with $M > 0$. Set $a_L = \frac{L}{8\pi}$. If $L > 16M$, then

$$N \geq a_L \cot\left(\frac{M}{a_L}\right) - 1. \quad (69)$$

In particular, for $L \geq \max(1, 8\pi\sqrt{M^2 + 2M})$,

$$N \geq \frac{L^2}{128\pi^2 M}, \quad \Omega := 2\pi \frac{N}{L} \geq \frac{L}{64\pi M}. \quad (70)$$

Along any such family with $L \rightarrow \infty$,

$$\liminf \frac{N}{L^2} \geq \frac{1}{64\pi^2 M}. \quad (71)$$

Proof. Every corrected-root contribution in (12) is nonnegative. The untouched lattice gives one weight per positive zero pair, so

$$M \geq \mu_{\text{lat}} = \sum_{n>N} \frac{a_L^2}{n^2 + a_L^2} \geq \int_{N+1}^{\infty} \frac{a_L^2}{t^2 + a_L^2} dt = a_L \arctan\left(\frac{a_L}{N+1}\right). \quad (72)$$

The integral comparison uses a decreasing positive function. If $L > 16M$, then $\frac{M}{a_L} < \frac{\pi}{2}$ and inversion of the tangent gives (69). For $0 < u < \frac{\pi}{2}$, the inequalities $\sin u \leq u$ and $\cos u \geq 1 - \frac{u^2}{2}$ imply $\cot u \geq \frac{1}{u} - \frac{u}{2}$. Hence

$$N \geq \frac{a_L^2}{M} - \frac{M}{2} - 1. \quad (73)$$

The stated threshold ensures $a_L^2 \geq M^2 + 2M$ and gives (70). Finally, $u \cot u \rightarrow 1$ as $u \rightarrow 0$ gives the lower limit. Thus $N \rightarrow \infty$ follows from the complete mean bound; it was not assumed in this calculation.

Remark (A surviving lattice factor). If $\frac{N}{L^2} \rightarrow c \in (0, \infty)$, the untouched factor in (11) satisfies, locally uniformly,

$$P_{\text{lat}}(z) \rightarrow \exp\left(-\frac{z^2 + \alpha^2}{4\pi^2 c}\right). \quad (74)$$

Indeed, putting $d_n = (\xi_n^2 + \alpha^2)^{-1}$, comparison with the reciprocal-square tail gives

$$\sum_{n>N} d_n \rightarrow \frac{1}{4\pi^2 c}, \quad \sum_{n>N} d_n^2 = O(L^4 N^{-3}) = O(L^{-2}). \quad (75)$$

On any fixed compact set, $\max_{n>N} |(z^2 + \alpha^2)d_n| = O(L^{-2})$. The logarithmic expansion of $\prod_{n>N} (1 - (z^2 + \alpha^2)d_n)$ proves (74). The limiting factor equals one at $i\alpha$ and has no zeros; its mean contribution is $\frac{1}{64\pi^2 c}$. Such a factor explains why zero retention does not by itself identify the limiting entire function.

A.3 Gaussian periodization and Fourier approximation

The forced resolution is useful because Gaussian tests have Gaussian Fourier tails. Fix one width $\sigma > 0$ and, for each real y , put

$$\psi_y(t) = \exp\left(-\frac{(t-y)^2}{2\sigma^2}\right), \quad \psi_{y,L}^{\text{per}}(t) = \sum_{k \in \mathbb{Z}} \psi_y(t + kL), \quad p = \Pi_N \psi_{y,L}^{\text{per}}. \quad (76)$$

All constants below may depend on the fixed translation y and width σ , but not on L, N or the selected state.

Lemma (Gaussian approximation). For $L \geq \max(1, 4|y|)$ and $\Omega \geq \max(1, \sigma^{-1})$,

$$\|p - \psi_y\|_{H^1(I)} \leq \Delta_{L,\Omega}, \quad (77)$$

where the Gaussian on the left is restricted to I and

$$\begin{aligned} \Delta_{L,\Omega} &:= C_\sigma \sqrt{L} e^{-\frac{L^2}{64\sigma^2}} \\ &\quad + \sqrt{\Omega + \frac{1 + (2\sigma^2)^{-1}}{\Omega}} e^{-\sigma^2 \frac{\Omega^2}{2}}. \end{aligned} \quad (78)$$

In particular, (70) makes this error Gaussian-small in L .

Proof. For $|t| \leq \frac{L}{2}$ and $k \neq 0$,

$$|t + kL - y| \geq \left(|k| - \frac{3}{4}\right)L \geq |k|\frac{L}{4}. \quad (79)$$

The bound

$$|\psi_y(u)| + |\psi'_y(u)| \leq \left(1 + \frac{\sqrt{\frac{2}{e}}}{\sigma}\right) e^{-\frac{(u-y)^2}{4\sigma^2}} \quad (80)$$

therefore gives

$$\|\psi_{y,L}^{\text{per}} - \psi_y\|_{H^1(I)} \leq C_\sigma \sqrt{L} e^{-\frac{L^2}{64\sigma^2}}. \quad (81)$$

The image series and its derivative converge uniformly. Unfolding its Fourier coefficient gives

$$\langle e_n, \psi_{y,L}^{\text{per}} \rangle = \sqrt{2\pi} \frac{\sigma}{\sqrt{L}} e^{-\sigma^2 \frac{\xi_n^2}{2}} e^{-iy\xi_n}. \quad (82)$$

Parseval, including the derivative, yields

$$\|p - \psi_{y,L}^{\text{per}}\|_{H^1(I)}^2 = 2\pi \frac{\sigma^2}{L} \sum_{|n| > N} (1 + \xi_n^2) e^{-\sigma^2 \xi_n^2}. \quad (83)$$

For the stated Ω , the summand decreases with the positive frequency. The mesh $2\frac{\pi}{L}$ and the two frequency signs give

$$\|p - \psi_{y,L}^{\text{per}}\|_{H^1(I)}^2 \leq 2\sigma^2 \int_{\Omega}^{\infty} (1 + u^2) e^{-\sigma^2 u^2} du \leq \left(\Omega + \frac{1 + (2\sigma^2)^{-1}}{\Omega}\right) e^{-\sigma^2 \Omega^2}. \quad (84)$$

The last inequality follows by integration by parts and

$$\int_{\Omega}^{\infty} e^{-\sigma^2 u^2} du \leq \frac{e^{-\sigma^2 \Omega^2}}{2\sigma^2 \Omega}. \quad (85)$$

Adding the image error proves the lemma. The displayed prefactors are polynomial; with $\Omega \geq \frac{L}{64\pi M}$ they can be absorbed into a weaker Gaussian exponent.

A.4 From finite residuals to Gaussian current tests

Lemma (Interval current and spatial comparison). The complete interval form has a representing operator $H_L : H^1(I) \rightarrow L^2(I)$ satisfying

$$\|H_L f\|_2 \leq A_L \|f\|_{H^1(I)}, \quad A_L = C_\Gamma + (4L + 2)e^{\frac{L}{2}}, \quad L \geq 1, \quad (86)$$

where C_Γ is absolute. For $r = \psi_y|_I$, $w = B_s h$ and $U = \mathcal{F}^{-1}(\varphi_s C)$,

$$|W(\psi_y, U) - W_L(r, w)| + |\langle \psi_y, U \rangle - \langle r, w \rangle| \leq C e^{-\kappa L^2}, \quad (87)$$

where one can take $\kappa = \frac{1}{64 \max(p_s, \sigma^2)}$ and $p_s = s + 2 + 16M$.

Proof. Write $S_q^L f(t) = \mathbf{1}_I(t + q)f(t + q)$, so $(S_q^L)^* = S_{-q}^L$. Polarization of (47) gives

$$\begin{aligned}
H_L f(t) &= a_0 f(t) + \int_0^\infty \rho(q) (2f - S_q^L f - S_{-q}^L f)(t) dq \\
&\quad - \sum_{1 < n < e^L} \frac{\Lambda(n)}{\sqrt{n}} (S_{\log n}^L f + S_{-\log n}^L f)(t) \\
&\quad + e^{\frac{t}{2}} \int_I e^{-\frac{u}{2}} f(u) du + e^{-\frac{t}{2}} \int_I e^{\frac{u}{2}} f(u) du.
\end{aligned} \tag{88}$$

The finite matrix is the compression of this representing operator: for $f, g \in V_N$, $\langle f, Hg \rangle = \langle f, H_L g \rangle = W_L(f, g)$.

For $0 < q \leq 1$ and $L \geq 1$, the overlap and killed endpoint strip give

$$\|f - S_q^L f\|_2 \leq q \|f'\|_2 + \sqrt{q} \|f\|_\infty, \quad \|f\|_\infty \leq \sqrt{2} \|f\|_{H^1(I)}. \tag{89}$$

The opposite orientation has the same bound. Since $\rho(q) \sim \frac{1}{2q}$ at zero, the subtracted Γ vector integral converges in L^2 . Its norm is bounded by $C_T \|f\|_{H^1}$, where one may take

$$C_T = |a_0| + 2 \int_0^1 \rho(q) (q + \sqrt{2q}) dq + 4 \int_1^\infty \rho(q) dq. \tag{90}$$

The killed prime shifts have norm at most one, and elementary counting gives

$$2 \sum_{1 < n < e^L} \frac{\Lambda(n)}{\sqrt{n}} \leq 4Le^{\frac{L}{2}}. \tag{91}$$

Each polar rank-one map has norm $2 \sinh(\frac{L}{2})$; their sum is bounded by $2e^{\frac{L}{2}}$. These are the cross terms whose quadratic form is the signed difference of the cosh and sinh terms in (47). This proves (86) independently of N .

For spatial transport, zero extension requires its endpoint jumps to be kept. If f is piecewise H^1 , with finitely many jumps, set

$$\mathfrak{b}(f) = \|f\|_2 + \|f'_{\text{pw}}\|_2 + \sum_x |[f]_x|. \tag{92}$$

Integrating its distributional derivative over an interval of length q gives

$$\|f(\cdot + q) - f\|_2 \leq q \|f'_{\text{pw}}\|_2 + \sqrt{q} \sum_x |[f]_x| \leq \sqrt{q} \mathfrak{b}(f), \quad 0 < q \leq 1. \tag{93}$$

Thus the polarized Γ form is bounded by $C \mathfrak{b}(f) \mathfrak{b}(g)$: near zero the product contributes $q\rho(q)$, and at infinity ρ is integrable. This does not treat the zero extensions as global H^1 functions.

For Gaussian envelopes, completing the square gives

$$|\langle f, g(\cdot + q) \rangle| \leq A_f A_g \sqrt{\frac{\pi}{2c}} e^{-c \frac{q^2}{2}} \quad \text{if} \quad |f(t)| \leq A_f e^{-ct^2}, \quad |g(t)| \leq A_g e^{-ct^2}. \tag{94}$$

The corresponding absolute prime sum converges using only $\Lambda(n) \leq \log n$. The two polar terms are bounded by their ordinary exponential moments.

Let r^0, w^0 be the zero extensions. The finite/global identity is

$$W_L(r, w) = W(r^0, w^0). \tag{95}$$

For prime delays $q < L$ the overlaps are exactly the killed overlaps. At $q = L$ their intersection has measure zero, and for $q > L$ it is empty. This also handles a prime power exactly equal

to e^L . The Γ form retains its diagonal tail for $q \geq L$, on both sides of the identity. Both polar moments are unchanged.

To estimate the difference from the whole-line Gaussian pair, use

$$(t + kL)^2 \geq \frac{t^2}{2} + k^2 \frac{L^2}{8}, \quad |t| \leq \frac{L}{2}, \quad k \neq 0. \quad (96)$$

With $c_U = \frac{1}{8p_s}$, the image sum, its derivative and the exterior bound give

$$|(U - w^0)(t)| + |(U - w^0)'_{\text{pw}}(t)| \leq C e^{-c_U \frac{L^2}{8}} e^{-c_U \frac{t^2}{2}}, \quad (97)$$

and the two jumps have total size at most $C e^{-c_U \frac{L^2}{4}}$. The cut error $\psi_y - r^0$ has analogous bounds with $c_\psi = \frac{1}{8\sigma^2}$; constants can depend on y . Apply the Γ jump estimate, the Gaussian prime-correlation bound and the polar moment bounds to these differences. They prove (87), including its inner-product term.

Lemma (Gaussian current passage). Along every subsequence with $C_j \rightarrow F$ locally uniformly and $E_j \rightarrow E$,

$$W(\psi_y, U_F) = E \langle \psi_y, U_F \rangle \quad \text{for every } y \in \mathbb{R}, \quad U_F = \mathcal{F}^{-1}(\varphi_s F). \quad (98)$$

Proof. Put $p_j = \Pi_{N_j} \psi_{y, L_j}^{\text{per}}$, $r_j = \psi_y|_{I_j}$ and $\Delta_j = \Delta_{L_j, \Omega_j}$. The preceding estimates give uniformly bounded H^1 norms for p_j, r_j, w_j and uniformly bounded form values $W_{L_j}(r_j, r_j)$. Using (86) for their difference and the bounded energies from A.1 yields

$$0 \leq \langle p_j, K_j p_j \rangle \leq C(1 + A_{L_j} \Delta_j). \quad (99)$$

Here positivity is used only for the vectors in V_{N_j} . Positive-form Cauchy–Schwarz gives

$$|\langle p_j, K_j w_j \rangle| \leq C(1 + A_{L_j} \Delta_j)^{\frac{1}{2}} \sqrt{\varepsilon_j}. \quad (100)$$

Replacing p_j by r_j costs at most $C(A_{L_j} + 1)\Delta_j$, since $\|H_{L_j} w_j\|_2 \leq A_{L_j} \|w_j\|_{H^1}$. Spatial comparison now gives

$$\begin{aligned} |W(\psi_y, U_j) - E_j \langle \psi_y, U_j \rangle| &\leq C \left[(1 + A_{L_j} \Delta_j)^{\frac{1}{2}} \sqrt{\varepsilon_j} \right. \\ &\quad \left. + (A_{L_j} + 1) \Delta_j + e^{-\kappa L_j^2} \right]. \end{aligned} \quad (101)$$

By (70) and (78), $A_{L_j} \Delta_j \rightarrow 0$. The Gaussian filtered-energy estimate (56) handles the remaining residual. Every term in (101) therefore tends to zero.

On the real frequency line, $\varphi_s C_j$ and each fixed polynomial multiple have a common Gaussian bound. Dominated inverse Fourier integrals give convergence of U_j and every derivative on compact physical sets. Their uniform spatial Gaussian bounds then give convergence in every fixed exponentially weighted L^1 and L^2 norm. The Γ pairings converge after local subtraction, the prime pairings through a summable Gaussian correlation bound, and the polar terms through their exponential moments. This proves (98). The subsequence was chosen only from C_j, E_j : for any fixed y the same estimate applies to it, so no test-dependent extraction is involved.

A.5 Gaussian uniqueness for the physical current

The preceding lemma supplies translated Gaussian tests of one width. We now identify the residual in its actual growth class, so that these tests determine a physical equation.

Lemma (Residual and uniqueness). The whole-line current of U_F is smooth and satisfies

$$\mathcal{H}U_F = EU_F \quad \text{pointwise on } \mathbb{R}. \quad (102)$$

Proof. Define $\mathcal{H}U$ by (88) with ordinary whole-line translations, all integer prime delays, and whole-line polar moments. For a smooth U with the Gaussian derivative bounds from A.1, the symmetric Γ difference is $O(q^2)$ at zero, after every fixed derivative in t . Its large- q kernel is integrable. The prime series and its derivatives converge locally normally. To bound them globally, group $\log n$ in $[m, m+1)$ and use

$$\sum_{e^m \leq n < e^{m+1}} \frac{\Lambda(n)}{\sqrt{n}} \leq e(m+1)e^{\frac{m}{2}}. \quad (103)$$

Completing the square in the resulting shifted Gaussian sum gives, for every fixed r ,

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} |U^{(r)}(t \pm \log n)| \leq C_r(1 + |t|)e^{\frac{|t|}{2}}. \quad (104)$$

The Γ part is bounded using U, U'' , and the polar part has the explicit envelope $Ce^{\frac{|t|}{2}}$. Thus the residual

$$T = \mathcal{H}U_F - EU_F \in C^\infty(\mathbb{R}), \quad |T(t)| \leq C(1 + |t|)e^{\frac{|t|}{2}} \quad (105)$$

is a function. These estimates also justify

$$W(\psi_y, U_F) - E\langle \psi_y, U_F \rangle = \int_{\mathbb{R}} \psi_y(t) T(t) dt. \quad (106)$$

Set $g(t) = T(t)e^{-\frac{t^2}{2\sigma^2}}$ and

$$G(z) = \int_{\mathbb{R}} g(t)e^{zt} dt. \quad (107)$$

The growth bound makes G entire, with differentiation dominated on every compact z -set. Equation (98) says

$$0 = e^{-\frac{y^2}{2\sigma^2}} G\left(\frac{y}{\sigma^2}\right) \quad (y \in \mathbb{R}). \quad (108)$$

The identity theorem gives $G = 0$. On the imaginary axis it is the Fourier transform of the L^1 function g , up to reversal of frequency. Fourier uniqueness gives $g = 0$ almost everywhere, hence $T = 0$ everywhere by continuity. For the specific uniqueness used here, one may convolve g with Gaussian approximate identities, apply Gaussian Fourier inversion and Fubini, and then pass to the L^1 limit. No bounded whole-line L^2 current or temperedness assertion enters this argument.

A.6 Dense θ tests and the limiting energy

Lemma (Dense radical tests). There are even θ -derived tests k_j whose span is dense in ordinary even $L^2(\mathbb{R})$ and for which $W(k_j, U_F) = 0$. They are admissible in (102) and force $E = 0$.

Proof. Define the θ trace

$$k_0(t) = e^{\frac{t}{2}} \sum_{n \geq 1} (4\pi^2 n^4 e^{4t} - 6\pi n^2 e^{2t}) e^{-\pi n^2 e^{2t}}. \quad (109)$$

To check evenness directly, put $h_0(u) = (4\pi^2 u^4 - 6\pi u^2)e^{-\pi u^2}$. Differentiating the Gaussian Fourier transform in the convention with phase $e^{-2\pi iuv}$ gives $\mathcal{F}h_0 = h_0$ and $h_0(0) = \int_{\mathbb{R}} h_0 = 0$. Poisson summation for its dilations therefore makes k_0 even. Mellin integration, initially in its convergent half-plane and then by evenness, gives $\hat{k}_0(z) = \Xi(z)$. More explicitly, the Mellin transform of $(4\pi^2 u^4 - 6\pi u^2)e^{-\pi u^2}$ is $s(s-1)\pi^{-\frac{s}{2}}\frac{\Gamma(\frac{s}{2})}{2}$; summing the integer dilations supplies $\zeta(s)$.

The trace is holomorphic in $|\Im t| < \frac{\pi}{4}$. On each closed smaller strip, its positive-end series and all derivatives are bounded by a polynomial in $e^{2|\Re t|}$ times $\exp(-ce^{2|\Re t|})$; evenness controls the other end. A contour shift consequently gives $|\Xi(t)| \leq C_\eta e^{-\eta|t|}$ for $0 < \eta < \frac{\pi}{4}$.

Put $k_j = (\partial_t^2 - \frac{1}{4})^j k_0$. Then

$$\hat{k}_j(z) = \left(-\left(z^2 + \frac{1}{4}\right)\right)^j \Xi(z). \quad (110)$$

We verify the noncompact test class directly before applying the explicit formula. For $f = k_j$ and $g = U_F$, set $\tilde{f}(t) = \overline{f(-t)}$, $c = \tilde{f} * g$ and $q(u) = u^{-\frac{1}{2}}c(\log u)$. Completing the square in the convolution bounds every derivative of c by a Gaussian. Thus q is smooth on $(0, \infty)$, and both q and

$$q'(u) = u^{-\frac{3}{2}}\left(c'(\log u) - \frac{1}{2}c(\log u)\right) \quad (111)$$

have arbitrarily strong endpoint power decay. In particular q belongs to the Weil class of [1, §3, (3.1)–(3.6)]. If $\rho = \frac{1}{2} + i\gamma$, its Mellin value is $\hat{c}(-\gamma)$, while

$$\hat{c}(z) = \overline{\hat{f}(\bar{z})}\hat{g}(z). \quad (112)$$

Reindexing the symmetric zero divisor gives exactly (57), with its reflected conjugation. Integration by parts gives rapid Fourier decay on $|\Im \gamma| \leq \frac{1}{2}$; the classical $O(T \log T)$ zero count makes the zero sum absolutely convergent. The arithmetic side converges absolutely by Gaussian prime correlations, exponential polar moments and the local Γ subtraction. These arguments also apply to the divided θ tests below.

The explicit formula now gives $W(k_j, U_F) = 0$, since each value at a ζ zero vanishes. Equation (102) can be paired directly with k_j : the envelope (105), and the same bound for $\mathcal{H}U_F$, are integrable against all θ -tail derivatives. The preceding absolute estimates justify each interchange with the current. Thus

$$0 = E\langle k_j, U_F \rangle \quad (j \geq 0). \quad (113)$$

The span of the k_j is dense in even $L^2(\mathbb{R})$. To prove this, let an even f be orthogonal to every k_j . By Plancherel, all even moments of $\hat{f}(t)\Xi(t)$ vanish. This function is in $L^1(e^{a|t|} dt)$ for every $a < \frac{\pi}{4}$, by Cauchy–Schwarz and the preceding Fourier decay. Its Fourier transform is holomorphic in a strip; all derivatives at zero vanish, with the odd ones vanishing by parity. The identity theorem and Fourier uniqueness imply $\hat{f}\Xi = 0$ almost everywhere. The real zeros of the nonzero entire function Ξ form a discrete set, so $f = 0$. Since U_F is even and nonzero, this density forces $E = 0$.

The topology in this last step is ordinary even L^2 : the continuous functional being tested is $f \mapsto E\langle f, U_F \rangle$. Each equality with $W(k_j, U_F)$ was justified separately by decay and the explicit formula. Density therefore eliminates the energy in this particular current equation; it does not extend W to a bounded form on L^2 or make W vanish on all even tests.

A.7 Divided θ tests and individual zeros

Lemma (Zero isolation). For each zero pair $\pm\gamma$ of Ξ , a legal even test can be chosen whose reflected Fourier values are nonzero at that pair and zero at every other zero. The weak current equation with $E = 0$ then forces $\widehat{U}_F(\gamma) = 0$.

Proof. Let $\gamma \neq 0$ be a zero of Ξ of order m . To isolate its value in the second slot of (57), we construct a legal test with transform

$$\Phi_\gamma(z) = \frac{\Xi(z)}{(z^2 - \bar{\gamma}^2)^m}. \quad (114)$$

For any smooth θ -tail function k with $\hat{k}(\gamma) = 0$, set

$$R_\gamma k(t) = ie^{i\gamma t} \int_{-\infty}^t e^{-i\gamma u} k(u) du = -ie^{i\gamma t} \int_t^\infty e^{-i\gamma u} k(u) du. \quad (115)$$

The vanishing moment makes the two expressions equal. Direct differentiation gives $(-i\partial_t - \gamma)R_\gamma k = k$, hence $\widehat{R_\gamma k}(z) = \frac{\hat{k}(z)}{z - \gamma}$.

Use the first integral on the negative tail and the second on the positive tail. For example,

$$\int_t^\infty \exp(|\Im \gamma|(u - t) - ce^{2u}) du \leq C_\gamma \exp(-c'e^{2t}). \quad (116)$$

The differential equation supplies the derivative bounds. These estimates persist on closed smaller physical strips: the moment there is still zero by a rectangular contour shift with vanishing vertical ends. Thus the divided function retains the requisite superexponential physical tails and exponential Fourier decay on bounded horizontal strips.

Conjugation preserves the zero multiplicity. Iterate this construction m times at $\bar{\gamma}$ and m times at $-\bar{\gamma}$. Every division uses a zero still present before that division. The final transform is (114), even, nonzero at $\pm\bar{\gamma}$, and zero at every other zero of Ξ . Its inverse transform is an admissible complex even test for (57) and (102). With $E = 0$, applying these equations gives

$$0 = 2m\overline{\Phi_\gamma(\bar{\gamma})}\widehat{U}_F(\gamma). \quad (117)$$

Here the reflected conjugation in the first slot makes the two surviving zero terms occur at $\pm\gamma$. The coefficient is nonzero because

$$\Phi_\gamma(\bar{\gamma}) = \frac{\Xi^{(m)}(\bar{\gamma})}{m!(2\bar{\gamma})^m} \neq 0. \quad (118)$$

Consequently $\widehat{U}_F(\gamma) = 0$ for each γ . The filter argument in Section 7 then gives $F(\gamma) = 0$.

This division removes a finite zero divisor only in a testing function. The explicit formula evaluates values, even at multiple zeros. The argument therefore proves value retention and makes no claim about the multiplicity of a zero of F .

B.1 Two odd sources and the complete mean

The mean condition can also be expressed by two inverse quadratic forms in the actual odd restriction. The same full least eigenvalue is used in both forms. This gives another expression of the compactness premise, rather than an estimate proving it.

Fix a section satisfying the hypotheses of Section 2. Let $V_\pm$ be its parity sectors, $P_\pm$ their orthogonal projections, and

$$B = K|_{V_-} = (H|_{V_-}) - EI > 0. \quad (119)$$

$$p_y = \Pi_N^- \left[\frac{\sinh(yt)}{y} \right], \quad \mathcal{J}(y) = \langle p_y, B^{-1} p_y \rangle \quad (y > 0), \quad (120)$$

where Π_N^- projects onto the odd Fourier sector. Put $\alpha = \frac{1}{4}$ and $\beta = \frac{3}{16}$.

Proposition (Two-source mean criterion). On any family of these same selected sections,

$$\sup_j \mu_j < \infty \quad \Leftrightarrow \quad \sup_j \frac{\mathcal{J}_j(\alpha)}{\mathcal{J}_j(\beta)} < \infty. \quad (121)$$

More precisely, with $\mathcal{R} = \frac{\mathcal{J}(\alpha)}{\mathcal{J}(\beta)}$,

$$\mathcal{R} \leq K_0, \quad K_0 \geq 1 \quad \Rightarrow \quad \mu \leq \frac{16}{7} \left[\frac{1}{2} \log K_0 + \log \left(\frac{16}{9} \right) \right], \quad (122)$$

while $\mu \leq M$ implies $\mathcal{R} \leq \left(\frac{16}{9} \right)^{2M}$.

Proof. We derive the finite block identity needed for the comparison. Write $K_+ = K|_{V_+}$, $H_+ = H|_{V_+}$, and put

$$d = P_+ D P_- : V_- \rightarrow V_+, \quad F = d^* d, \quad G = d^* K_+ d, \quad \zeta = d^* \delta, \quad (123)$$

$$\delta_0 = L^{-\frac{1}{2}}, \quad g = \delta_0^{-1} d^* H_+ e_0, \quad w = B^{-1} g. \quad (124)$$

The endpoint vector δ is even. The odd-even block of (6) is

$$d^* K_+ - B d^* = b \delta^*. \quad (125)$$

Its application to e_0 gives $g = b$, since $D e_0 = 0$. Multiplying on the right by d , and also applying the same block equation to the endpoint-normalized ground vector v , gives

$$G = B F + g \zeta^*, \quad [B, F] = \zeta g^* - g \zeta^*, \quad w = -d^* v. \quad (126)$$

The commutator identity is the difference of the first equality and its adjoint. The odd block of the squared corrected derivative is therefore

$$D_c^2|_{V_-} = F + w \zeta^* = B^{-1} G. \quad (127)$$

The quotient construction of Section 3, and the nonzero real spectrum proved in Section 2, imply that $B^{-\frac{1}{2}} G B^{-\frac{1}{2}}$ has eigenvalues $\eta_k^2 > 0$, once each. In particular $G > 0$; this conclusion uses the quotient spectrum, not merely $B > 0$.

For $u > 0$ set

$$R_u = (F + uI)^{-1}, \quad M_u = G + uB, \quad (128)$$

$$\mathfrak{d}(u) = \frac{\det M_u}{\det B \det(F + uI)} = 1 + \zeta^* R_u w = \prod_{k=1}^N \frac{\eta_k^2 + u}{\xi_k^2 + u} > 0. \quad (129)$$

This is the determinant lemma applied to (126). Combining it with the full sine factor in (8) and (10) yields

$$C(iy) = \frac{\alpha \sinh(y \frac{L}{2})}{y \sinh(\alpha \frac{L}{2})} \frac{\mathfrak{d}(y^2)}{\mathfrak{d}(\alpha^2)}. \quad (130)$$

In particular the untouched Fourier lattice has not been removed.

Put $\chi(u) = \zeta^* M_u^{-1} \zeta$. The adjoint form of (126) is

$$M_u = (F + uI)B + \zeta g^*. \quad (131)$$

It gives the ordered inverse identity

$$M_u^{-1} \zeta = \frac{B^{-1} R_u \zeta}{\mathfrak{d}(u)}. \quad (132)$$

Indeed, multiplying its numerator by M_u gives

$$\zeta[1 + g^* B^{-1} R_u \zeta] = \mathfrak{d}(u) \zeta. \quad (133)$$

The scalar in brackets is the conjugate of $1 + \zeta^* R_u w$, which is real by (129). No commutation of B and F is used. Differentiating with the matrices fixed now gives

$$-\chi'(u) = \frac{\langle R_u \zeta, B^{-1} R_u \zeta \rangle}{\mathfrak{d}(u)^2}. \quad (134)$$

On the other hand, the spectral theorem for $B^{-\frac{1}{2}} G B^{-\frac{1}{2}}$ gives

$$\chi(u) = \sum_{k=1}^N \frac{a_k}{\eta_k^2 + u}, \quad a_k \geq 0, \quad \sum_k a_k > 0. \quad (135)$$

The positive total follows because $\zeta \neq 0$ when $N \geq 1$.

In the orthonormal odd basis $O_k(t) = i\sqrt{\frac{2}{L}} \sin(\xi_k t)$, direct integration gives

$$\langle O_k, p_y \rangle = \frac{2i\sqrt{\frac{2}{L}}(-1)^k \xi_k \sinh(y\frac{L}{2})}{y(\xi_k^2 + y^2)}. \quad (136)$$

The coordinates of ζ are $\sqrt{\frac{2}{L}}(-1)^k \xi_k$. Hence

$$p_y = \frac{2i \sinh(y\frac{L}{2})}{y} R_{y^2} \zeta. \quad (137)$$

Equations (134) and (137) show that $\mathcal{J}(y) > 0$. Substitution into (130) cancels the hyperbolic factors and proves

$$C(iy)^2 = \frac{\mathcal{J}(y) (-\chi'(\alpha^2))}{\mathcal{J}(\alpha) (-\chi'(y^2))}. \quad (138)$$

At $y = \beta$, the last ratio is a positive weighted average of

$$\left[\frac{\eta_k^2 + \beta^2}{\eta_k^2 + \alpha^2} \right]^2, \quad (139)$$

so it lies between $(\frac{\beta}{\alpha})^4 = \frac{81}{256}$ and 1. Taking logarithms in (138) yields

$$\frac{1}{2} \log \mathcal{R} \leq -\log C(i\beta) \leq \frac{1}{2} \log \mathcal{R} + \log\left(\frac{16}{9}\right). \quad (140)$$

Combining this with the complete-product estimate (38) gives (122) and its converse bound. The ratio $\mathcal{R}$ need not be at least one; $K_0 \geq 1$ is simply a convenient chosen upper bound. $\square$

Both inverse problems in (120) use the same actual positive odd block, with the full least energy subtracted. The proposition leaves the uniform ratio bound and the existence of an unbounded selected arithmetic family unresolved, exactly as in the mean formulation of the moving theorem.

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